

Cambridge (CIE) IGCSE Additional Maths



Your notes

Factors of Polynomials

Contents

- * Operations with Polynomials
- * Polynomial Division
- * Factor & Remainder Theorem
- * Solving Cubic Equations



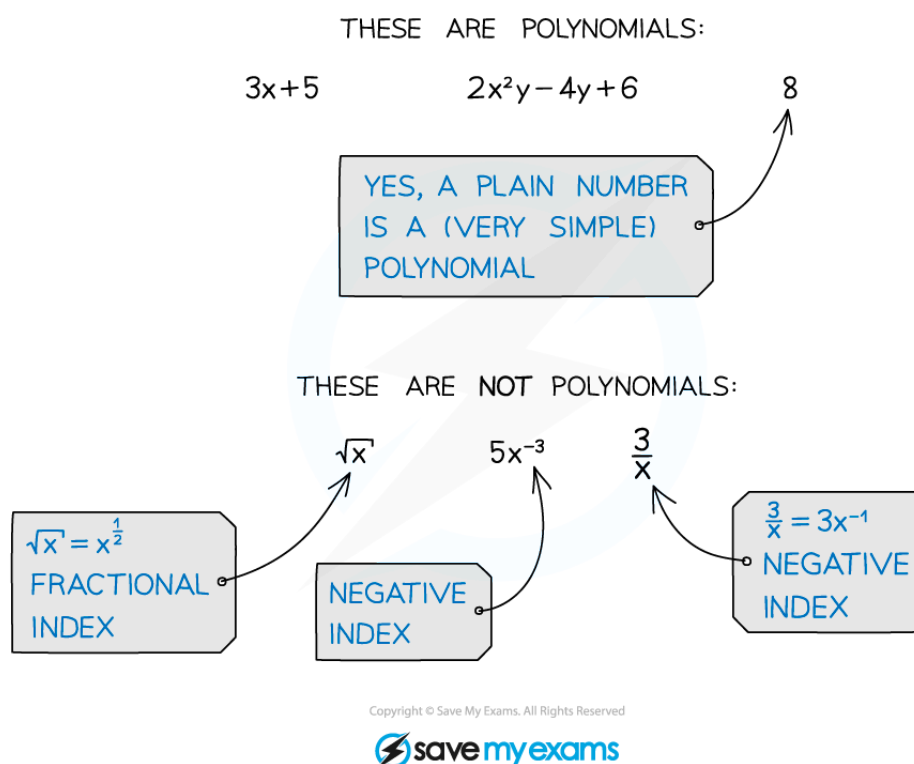
Your notes

Operations with Polynomials

Expanding Brackets

What is a polynomial?

- A **polynomial** is an algebraic expression consisting of a finite number of terms, with **non-negative integer** indices only



What do I need to know about expanding brackets?

- Expanding brackets** is now extended to include
 - brackets with more than 2 terms
 - brackets that are **polynomials** (beyond linear)



Your notes

- To expand two brackets, **each term** in the **first** set of brackets must be **multiplied** by **each term** in the **second** set of brackets
 - If there are more than 2 brackets to expand, only work with a pair at a time
 - You may have seen or used FOIL to expand quadratic expressions
 - that is helpful but is a special case (shortcut) of this more general approach

$$(a + b)(x + y + z) = a(x + y + z) + b(x + y + z)$$

$$= ax + ay + az + bx + by + bz$$

$$(a + b + c)(x + y + z) = a(x + y + z) + b(x + y + z) + c(x + y + z)$$

$$= ax + ay + az + bx + by + bz + cx + cy + cz$$

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- If you are trying to expand a two term bracket to a higher power (i.e. $(a + b)^n$) then use the **binomial expansion**



Worked Example



Your notes

Expand $(x - 4)(x + 3)(x^2 + x - 1)$

FIRST EXPAND

$$(x - 4)(x + 3) = x^2 - x - 12$$

THEN CONTINUE:

$$(x^2 - x - 12)(x^2 + x - 1)$$

$$= x^2(x^2 + x - 1) - x(x^2 + x - 1) - 12(x^2 + x - 1)$$

$$= x^4 + x^3 - x^2 - x^3 - x^2 + x - 12x^2 - 12x + 12$$

$$= x^4 - 14x^2 - 11x + 12$$

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Worked Example



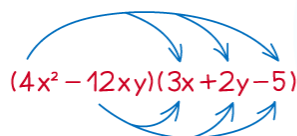
Your notes



Expand and simplify $4x(x - 3y)(3x + 2y - 5)$.

EXPAND $4x(x - 3y)$ FIRST

$$4x(x - 3y) = 4x^2 - 12xy$$

$$(4x^2 - 12xy)(3x + 2y - 5)$$


MULTIPLY EACH TERM IN THE FIRST BRACKET BY EACH TERM IN THE SECOND BRACKET

$$= 12x^3 + 8x^2y - 20x^2 - 36x^2y - 24xy^2 + 60xy$$

$$= 12x^3 - 28x^2y - 20x^2 - 24xy^2 + 60xy$$

COLLECT LIKE TERMS

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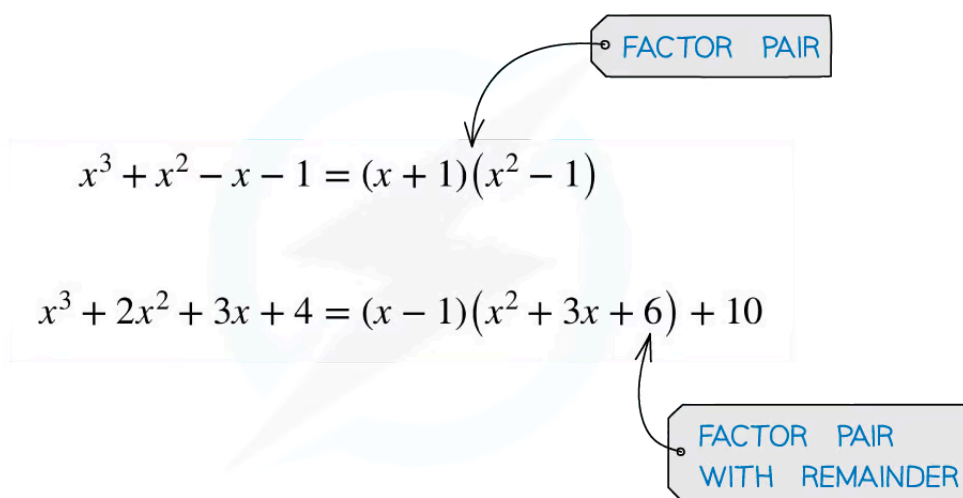
Your notes

Polynomial Division

Polynomial Division

What is polynomial division?

- Polynomial division is the process of **dividing two polynomials**
 - This is usually only useful when the **degree of the denominator** is **less than or equal** to the **degree of the numerator**
- Polynomial division** is a method for splitting polynomials into factor pairs
 - (with or without a remainder term)



The diagram illustrates two types of polynomial division results. The first equation, $x^3 + x^2 - x - 1 = (x + 1)(x^2 - 1)$, is labeled with a callout box 'FACTOR PAIR' pointing to the right-hand side. The second equation, $x^3 + 2x^2 + 3x + 4 = (x - 1)(x^2 + 3x + 6) + 10$, is labeled with a callout box 'FACTOR PAIR WITH REMAINDER' pointing to the right-hand side. A large, faint lightning bolt watermark is visible in the background of the equations.

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- The main uses of polynomial division are
 - factorising polynomials
 - simplifying 'top-heavy' algebraic fractions

How do I divide polynomials?

- The method used for polynomial division is just like the **long division** method for numbers
 - sometimes called the 'bus stop method'



Your notes

$$\begin{array}{r}
 124 \\
 39 \overline{) 4836} \\
 \underline{-39} \\
 936 \\
 \underline{-78} \\
 156 \\
 \underline{-156} \\
 0
 \end{array}$$

Annotations for the long division:

- 1×39 (points to the first subtraction)
- 2×39 (points to the second subtraction)
- 4×39 (points to the third subtraction)

so $\frac{4836}{39} = 124$

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- The answer to a polynomial division question is built up term by term
 - Start by dividing by the highest power term
 - Write out this multiplied by the divisor and subtract



Your notes



Divide the polynomial $f(x) = x^3 + 6x^2 - 9x - 14$ by $(x - 2)$.

DEAL WITH HIGHEST POWER TERM OF THE DIVIDEND FIRST, AND COMPARE HIGHEST POWER TERM OF THE DIVISOR. x^2 TIMES x EQUALS x^3 , SO x^2 GOES HERE.

$$\begin{array}{r} x^2 \\ (x-2) \overline{) x^3 + 6x^2 - 9x - 14} \\ \underline{-(x^3 - 2x^2)} \\ 8x^2 - 9x - 14 \end{array}$$

x^2 TIMES $x-2$ EQUALS $x^3 - 2x^2$, SO SUBTRACT THAT HERE

$$\begin{aligned} x^3 - x^3 &= 0 \\ 6x^2 - (-2x^2) &= 8x^2 \end{aligned}$$

YOU HAVEN'T DONE ANYTHING TO $-9x - 14$, SO CARRY THAT DOWN

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- Continue to divide by each reducing power term and subtracting your answer each time



Your notes

NEXT DEAL WITH THE x^2 TERM IN YOUR CURRENT REMAINDER.
 $8x$ TIMES x EQUALS $8x^2$, SO $+8x$ GOES HERE AND WE PROCEED AS BEFORE.

$$\begin{array}{r}
 x^2 + 8x \\
 \hline
 (x-2) \overline{) x^3 + 6x^2 - 9x - 14} \\
 \underline{-(x^3 - 2x^2)} \\
 8x^2 - 9x - 14 \\
 \underline{-(8x^2 - 16x)} \\
 7x - 14
 \end{array}$$

$8x(x-2) = 8x^2 - 16x$

$8x^2 - 9x - 14 - (8x^2 - 16x) = 7x - 14$

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- Continue until you are left with zero

AND REPEAT FOR THE x TERM OF THE CURRENT REMAINDER.
7 TIMES x EQUALS $7x$, SO $+7$ GOES HERE.

$$\begin{array}{r}
 x^2 + 8x + 7 \\
 \hline
 (x-2) \overline{) x^3 + 6x^2 - 9x - 14} \\
 \underline{-(x^3 - 2x^2)} \\
 8x^2 - 9x - 14 \\
 \underline{-(8x^2 - 16x)} \\
 7x - 14 \\
 \underline{-(7x - 14)} \\
 0
 \end{array}$$

$7(x-2) = (7x-14)$

$x-2$ IS A FACTOR OF $x^3 + 6x^2 - 9x - 14$,
SO THERE IS NO REMAINDER TERM
(i.e. THE REMAINDER IS ZERO)

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- If the divisor is **not** a **factor** of the **polynomial** then there will be a **remainder** term left at the end of the division



Worked Example

For the polynomial $f(x) = x^4 + 11x^2 - 1$ divide $f(x)$ by $x + 3$ and write the remainder.

Set up the polynomial division ('bus stop')

There is no x^3 term so write this as $0x^3$ in the method. There is no x term so write this as $0x$ in the method.

The first division step to consider is $x^4 \div x$.



Your notes

$$\begin{array}{r} x^3 \\ x+3 \overline{) x^4 + 0x^3 + 11x^2 + 0x - 1} \end{array}$$

.Multiply x^3 by $x+3$ and subtract from $x^4 + 0x$.

$$\begin{array}{r} x^3 \\ x+3 \overline{) x^4 + 0x^3 + 11x^2 + 0x - 1} \\ \underline{-(x^4 + 3x^3)} \\ -3x^3 \end{array}$$

Bring the $11x^2$ down and divide $-3x^3$ by x . Continue with each step until you are finished.

$$\begin{array}{r} x^3 - 3x^2 + 20x - 60 \\ x+3 \overline{) x^4 + 0x^3 + 11x^2 + 0x - 1} \\ \underline{-(x^4 + 3x^3)} \\ -3x^3 + 11x^2 \\ \underline{-(-3x^3 - 9x^2)} \\ 20x^2 + 0x \\ \underline{-(20x^2 + 60x)} \\ -60x - 1 \\ \underline{-(-60x - 180)} \\ 179 \end{array}$$

The remainder is 179.

Quadratic Divisor



Your notes

What is meant by a quadratic divisor?

- **Polynomial** division usually involves **dividing** by a **linear** term
 - a term of the form $(x + p)$ where p is a constant and usually an **integer**
- It is possible to divide a polynomial by a **quadratic term** (and cubic, etc)
 - this would be a term of the form $(x^2 + qx + r)$ where q and r are constants
 - this is what is meant by a **quadratic divisor**

How do I divide by a quadratic divisor?

- The process is the same as for a linear divisor
 - However, as x^2 will not divide into x (in the polynomial division sense at least) the **remainder**, if there is one, could be of **linear form**, i.e. $(rx + s)$ where r and s are constants
 - It is possible that $r = 0$ and so the remainder is still a constant



Examiner Tips and Tricks

- Give yourself plenty of room to do polynomial division
 - Not only will this help avoid errors, it will make your working clear
- If you make a mistake and change something, fine, but if your method starts to get too messy it is best to restart



Worked Example

Find the remainder when $x^4 + 4x^3 - x + 1$ is divided by $x^2 - 2x$.

Set up the polynomial division ('bus stop') - there is no x^2 term so write this as $0x^2$ in the method.

The first division step to consider is $x^4 \div x^2$.



Your notes

$$\begin{array}{r}
 \overline{x^2 + 6x + 12} \\
 x^2 - 2x \overline{) x^4 + 4x^3 + 0x^2 - x + 1} \\
 \underline{-(x^4 - 2x^3)} \\
 6x^3 + 0x^2 \\
 \underline{-(6x^3 - 12x^2)} \\
 12x^2 - x \\
 \underline{-(12x^2 - 24x)} \\
 23x + 1
 \end{array}$$

The remainder is $23x + 1$.



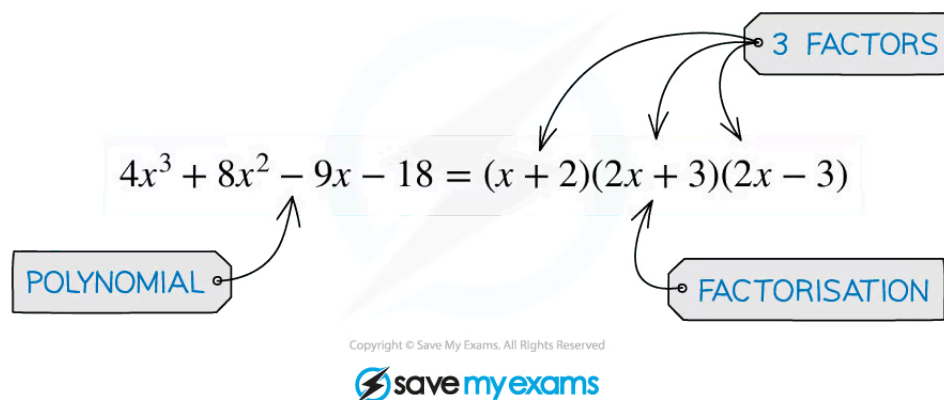
Your notes

Factor & Remainder Theorem

Factor Theorem

What is the factor theorem?

- The **factor theorem** is a useful result concerning the **roots** and **factors** of polynomials
 - In the example below, the polynomial $4x^3 + 8x^2 - 9x - 18$ has three (linear) **factors**
 - $(x + 2)$, $(2x + 3)$ and $(2x - 3)$
 - and so it has the three **roots** $x = -2$, $x = -\frac{3}{2}$ and $x = \frac{3}{2}$



- For a polynomial $f(x)$ the **factor theorem** states that:

i) if $f(p) = 0$, then $(x - p)$ is a factor of $f(x)$
 ($x = p$ is a root of $f(x)$)

and

ii) if $(x - p)$ is a factor of $f(x)$, then $f(p) = 0$



Your notes

$$g(x) = 2x^3 - 7x^2 + 9$$

$$\begin{aligned} g(3) &= 2(3)^3 - 7(3)^2 + 9 \\ &= 54 - 63 + 9 \\ &= 0 \end{aligned}$$

$$g(3) = 0$$

BY (i) THE FACTOR THEOREM
TELLS US THAT $(x-3)$ IS A
FACTOR OF $g(x)$

$$\begin{aligned} h(x) &= 4x^3 + 8x^2 - 9x - 18 \\ &= (x+2)(4x^2 - 9) \end{aligned}$$

$$(x+2) = (x - (-2))$$

IS A FACTOR

BY (ii) THE FACTOR THEOREM
TELLS US THAT $h(-2) = 0$
(TEST IT OUT!)

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Examiner Tips and Tricks

- In an exam, the values of p you'll need to find that make $f(p) = 0$ are going to be integers close to zero
 - Try $p = 1$ and $p = -1$ first, then 2 and -2, then 3 and -3
 - It is unlikely that you'll have to go beyond that



Worked Example

a) Show that $(x-2)$ is a factor of the polynomial $f(x) = x^3 + 6x^2 - 9x - 14$.



Your notes

(From part (ii) of our definition of factor theorem ...) ... if $(x - 2)$ is a factor of $f(x)$ then $f(2) = 0$.

$$f(2) = (2)^3 + 6(2)^2 - 9(2) - 14$$

$$f(2) = 8 + 24 - 18 - 14$$

$$f(2) = 0$$

Since $f(2) = 0$, $(x - 2)$ is factor of $f(x)$.

b) Use the factor theorem to find another factor of $f(x)$.

Try $f(1)$ first,

$$f(1) = (1)^3 + 6(1)^2 - 9(1) - 14$$

$$f(1) = 1 + 6 - 9 - 14$$

$$f(1) = -16$$

Since $f(1) \neq 0$, $(x - 1)$ is **not** a factor of $f(x)$.

Try $f(-1)$,

$$f(-1) = (-1)^3 + 6(-1)^2 - 9(-1) - 14$$

$$f(-1) = -1 + 6 + 9 - 14$$

$$f(-1) = 0$$

Since $f(-1) = 0$, $(x + 1)$ is a factor of $f(x)$.

$(x + 1)$ is another factor of $f(x)$.

$(x + 7)$ is the third (linear) factor.

Once one factor is known, polynomial division could be used to find the others. (In this case we were specifically asked to use factor theorem.)

Remainder Theorem

What is the remainder theorem?

- The **factor theorem** is actually a special case of the more general **remainder theorem**
- The **remainder theorem** states that when the polynomial $f(x)$ is divided by $(x - a)$ the remainder is $f(a)$
 - You may see this written formally as $f(x) = (x - a)Q(x) + f(a)$



Your notes

- In **polynomial division**
 - $Q(x)$ would be the **result** (at the top) of the division (the **quotient**)
 - $f(a)$ would be the **remainder** (at the bottom)
 - $(x - a)$ is called the **divisor**
- In the case when $f(a) = 0$, $f(x) = (x - a)Q(x)$ and hence $(x - a)$ is a factor of $f(x)$ - the **factor theorem**!

How do I solve problems involving the remainder theorem?

- If it is the **remainder** that is of particular interest, the remainder theorem saves the need to carry out polynomial division in full
 - e.g. The remainder from $(x^2 - 2x) \div (x - 3)$ is $3^2 - 2 \times 3 = 3$
 - This is because if $f(x) = x^2 - 2x$ and $a = 3$
- If the remainder from a polynomial division is known, the **remainder theorem** can be used to find unknown **coefficients** in polynomials
 - g. The remainder from $(x^2 + px) \div (x - 2)$ is 8 so the value of p can be found by solving $2^2 + p(2) = 8$, leading to $p = 2$
 - In harder problems there may be more than one unknown in which case **simultaneous equations** would need setting up and solving
- The more general version of remainder theorem is if $f(x)$ is divided by $(ax - b)$ then the remainder is $f\left(\frac{b}{a}\right)$
 - The remainder is still found by evaluating the polynomial at the value of x such that $ax - b = 0$ (the divisor is zero) but it is not necessarily an integer



Examiner Tips and Tricks

- Exam questions will use formal mathematical language which can make factor and remainder theorem questions sound more complicated than they are
 - Ensure you are familiar with the various terms from these revision notes



Worked Example

The polynomial $p(x)$ is given by $8x^4 + ax^2 + bx - 1$, where a and b are integer constants.

When $p(x)$ is divided by $(x - 1)$ the remainder is 9.

When $p(x)$ is divided by $(2x - 1)$ the remainder is 1.

Find the values of a and b .

Remainder theorem: " $f(a)$ is the remainder when $f(x)$ is divided by $(x - a)$ ".

$x - 1 = 0$ when $x = 1$:

$$\begin{aligned} p(1) &= 9 \\ 8 + a + b - 1 &= 9 \\ a + b &= 2 \end{aligned}$$

$2x - 1 = 0$ when $x = \frac{1}{2}$:

$$\begin{aligned} p\left(\frac{1}{2}\right) &= 1 \\ 8\left(\frac{1}{2}\right)^4 + a\left(\frac{1}{2}\right)^2 + b\left(\frac{1}{2}\right) - 1 &= 1 \\ \frac{1}{2} + \frac{1}{4}a + \frac{1}{2}b - 1 &= 1 \\ a + 2b &= 6 \end{aligned}$$

Solving simultaneously,

$$\begin{array}{r} a + b = 2 \\ a + 2b = 6 \\ \hline b = 4 \\ \therefore a = 2 - 4 = -2 \end{array}$$

$$a = -2, b = 4$$

$$p(x) = 8x^4 - 2x^2 + 4x - 1$$



Your notes



Your notes

Solving Cubic Equations

Solving Cubic Equations

What is a cubic equation?

- A **cubic** function is an **polynomial** of degree 3
 - i.e. the highest power of x is 3
- A cubic equation can be written in the form
 - $ax^3 + bx^2 + cx + d = 0$
- Solving a cubic equation involves **factorising** the cubic function first.

How do I factorise a cubic function?

- **Factorising** a cubic (function) combines the **factor theorem** with the method of **polynomial division**
- The example below shows the steps for factorising a cubic



Fully factorise $x^3 + 4x^2 - 11x - 30$.

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STEP 1

Use factor theorem.

Find a value p such that $f(p) = 0$.



Your notes

$$(1)^3 + 4(1)^2 - 11(1) - 30 = -36$$

TRY $x=1$

$$(-1)^3 + 4(-1)^2 - 11(-1) - 30 = -16$$

TRY $x=-1$

$$(2)^3 + 4(2)^2 - 11(2) - 30 = -28$$

TRY $x=2$

$$(-2)^3 + 4(-2)^2 - 11(-2) - 30 = 0 \checkmark$$

TRY $x=-2$

BY THE FACTOR THEOREM THIS TELLS US THAT $(x - (-2)) = (x + 2)$ IS A FACTOR OF THE POLYNOMIAL

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STEP 2

Use polynomial division.

Divide $f(x)$ by $(x - p)$.

(It is possible to do this step 'by inspection', see the worked example below)

$$\begin{array}{r}
 x^2 + 2x - 15 \\
 x+2 \overline{) x^3 + 4x^2 - 11x - 30} \\
 \underline{-(x^3 + 2x^2)} \\
 2x^2 - 11x - 30 \\
 \underline{-(2x^2 + 4x)} \\
 -15x - 30 \\
 \underline{-(-15x - 30)} \\
 0
 \end{array}$$

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STEP 3

Use the result of your division to write $f(x) = (x - p)(ax^2 + bx + c)$.



Your notes

STEP 4

If the quadratic $(ax^2 + bx + c)$ can be factorised, do so.

$f(x)$ can then be written as the product of three linear factors.

If the quadratic cannot be factorised, then the result from STEP 3 is the final factorisation.

$$x^2 + 2x - 15 = (x + 5)(x - 3)$$

$$\text{SO } x^3 + 4x^2 - 11x - 30 = (x + 2)(x + 5)(x - 3)$$

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How do I solve a cubic equation?

- A cubic equation will have either 1, 2 or 3 (real) solutions
 - (The cubic function will have either 1, 2 or 3 (real) roots)
- Once the cubic function is factorised using the four steps above, there is one more step to carry out

STEP 5 Find the solutions to the cubic equation by making each factor equal to zero

- For each **linear** factor, $(x - p)$
 - $x - p = 0$
 - so $x = p$ is a solution
 - This is the factor theorem!
- For a **quadratic** factor, $(ax^2 + bx + c)$
 - $ax^2 + bx + c = 0$
 - use either the **quadratic formula** or **completing the square** (as it won't factorise)
 - this will give two of the solutions to the cubic equation
 - if there are no solutions to the quadratic equation there are no solutions other than that from the linear factor
- From the example above,
 - $x^3 + 4x^2 - 11x - 30 = (x + 2)(x + 5)(x - 3)$



Your notes

- so the solutions to the cubic equation $x^3 + 4x^2 - 11x - 30 = 0$ are
 - $x = -2$, $x = -5$ and $x = 3$
- Cubic equations can have equal (repeated) solutions
 - e.g. $(x - 2)^2(x + 1)$ has two (equal and real) roots, $x = 2$ (repeated) and $x = -1$
 - e.g. $(2x - 1)^3$ has three (equal and real) roots, $x = \frac{1}{2}$



Examiner Tips and Tricks

- When $d = 0$ (i.e. there is no constant term) then x is a factor of the cubic function, and so $x = 0$ is a solution
 - This is a special case of factor theorem, where $f(0) = 0$
 - spotting the factor of x means there is no need to test values
 - Take out a factor of x and a quadratic function will remain
 - Deal with the quadratic in any of the usual ways



Worked Example

a) Solve the cubic equation $x^3 - 10x^2 + 12x + 8 = 0$.

STEP 1 - use factor theorem with $f(x) = x^3 - 10x^2 + 12x + 8$

$$f(1) = (1)^3 - 10(1)^2 + 12(1) + 8 = 11 \neq 0$$

$$f(-1) = (-1)^3 - 10(-1)^2 + 12(-1) + 8 = -15 \neq 0$$

$$f(2) = (2)^3 - 10(2)^2 + 12(2) + 8 = 0$$

$$\therefore (x - 2) \text{ is a factor of } f(x)$$

STEP 2 - polynomial division ($f(x) \div (x - 2)$) or 'by inspection' By inspection ...

$$f(x) = (x - 2)(ax^2 + bx + c)$$



Your notes

('cubic' $\div$ 'linear' = 'quadratic')

$$a = 1$$

(because the x^3 is generated **only** from $x \times ax^2$)

$$c = -4$$

(because the constant term is generated **only** from $-2 \times c$)Equate coefficients of x (or x^2) terms to find b ,

$$12 = c - 2b$$

$$b = \frac{-4 - 12}{2} = -8$$

STEP 3

$$f(x) = (x - 2)(x^2 - 8x - 4)$$

STEP 4 - the quadratic does not factorise

STEP 5 - Use the factors to find the solutions

$$x - 2 = 0, \quad x = 2$$

$$x^2 - 8x - 4 = 0, \quad x = \frac{8 \pm \sqrt{64 + 16}}{2}, \quad x = 4 \pm 2\sqrt{5}$$

(using the quadratic formula)

The solutions to $x^3 - 10x^2 + 12x + 8 = 0$ are $x = 2$, $x = 4 + 2\sqrt{5}$ and $x = 4 - 2\sqrt{5}$.